

EDUCATION

Master Degree in Computer Science and AI

Università degli Studi di Milano
2022 – 2026

Bachelor's Degree in Computer Science

Università degli Studi di Ferrara
2018 – 2021

SKILLS

AI Agents & LLMs

Prompt Engineering
AI Agents, Agentic Workflows
RAG, Vector Retrieval (PGVector)
VLMs, QLoRA
Quantization (FP16 / INT8)
PyTorch, Distributed Training
OpenCV, SAM, MediaPipe

Backend & Infrastructure

FastAPI, Async/Await
RabbitMQ, Celery
Docker
PostgreSQL + PGVector
Gradio

3D & Vision

CUDA
Gaussian Splatting, COLMAP
Open3D, BlenderProc
Unity, OpenXR

Programming Languages

Proficient in
C, C#, Python
Familiar with
C++, Java, Bash

Languages Spoken

Italian (native)
English (C1)

GIORGIA CARBONI

AI / Computer Vision Engineer

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PROFILE

AI engineer with hands-on experience building production-grade **agentic systems**, **RAG pipelines**, and scalable **backend AI infrastructure**. Skilled in fine-tuning and deploying **vision-language models** on consumer hardware, and in applying systematic **prompt engineering** to improve **LLM reliability** in production. Strong **product mindset** with experience embedding AI across diverse operational domains, from industrial inspection to Mixed Reality assistant services. Available to start immediately.

EXPERIENCE

AI Engineer

Fifth Ingenium | Milan, May 2025 – May 2026

Agentic AI Services — Designed and built a microservices backend (**FastAPI**, **RabbitMQ**, **Celery**) to orchestrate **LLM inference queues** and manage **agentic workflows** for a Mixed Reality assistant.

Prompt Engineering — Applied systematic **prompt engineering** across production LLM pipelines to improve **output reliability**, reduce hallucinations, and enable structured **multi-step reasoning**.

RAG Pipeline Development — Designed novel text extraction and chunking strategies for efficient **vector storage** and **LLM retrieval**; optimized token usage and embedding strategy to significantly reduce **API costs**.

Fine-tuned Vision-Language Models — Created domain-specific datasets and fine-tuned **VLMs** using **QLoRA** to bridge the consumer/industrial domain gap, reducing memory footprint by **2.5x** for consumer GPU deployment.

Synthetic Data Pipeline — Built automated pipelines for **training data generation** and annotation in industrial settings; pipeline presented at **MECSPE 2026**.

Joint 2D-3D Embedding Alignment — Orchestrated a **PyTorch** framework mapping **CLIP** (2D) and **Sonata** (3D) inputs into a shared latent space for cross-modal performance analysis; submitted to **SIGGRAPH LA 2026**.

Cloud Engineer

DXC Technology | April 2022 – April 2023

Maintained Linux-based virtualization infrastructure and implemented automation scripts for service monitoring.

MAIN PROJECTS

Atrial Fibrillation Detector — Python — **ECG signal processing** for early AF detection.
github.com/Giorgia01carboni/atrial-fibrillation-detector

GPU Optical Flow — **CUDA** — Pyramidal multi-scale optical flow pipeline for **real-time** video processing.

GPU Ray-Casting of Quadric Surfaces — C++, OpenGL — Real-time ray-casting of implicit quadratic surfaces. github.com/Giorgia01carboni/GPU-RayCasting-Quadrics

Ping Pong AI – Reinforcement Learning — C#, Unity — **RL agent** trained via self-play.
github.com/Giorgia01carboni/pAIingPong

Thesis — Python, C++, COLMAP — High-fidelity **3D Gaussian Splatting** and **SfM pipeline** optimised for consumer hardware, with **CUDA kernel optimisations** for real-time rasterisation and temporal super-sampling for dynamic scenes.